Pathway-Informed Deep Learning for Multi-Phenotype Therapy-Response Stratification in TCGA-BRCA

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Code and reproducible artifacts: https://github.com/sujoy166/TherapAgent

Abstract—Predicting whether a breast-cancer patient will receive targeted molecular therapy, radiation therapy, and survive at least six months is intrinsically a multi-label clinical decision driven by the joint activity of dozens of biological pathways — many of them immunological. We benchmark three recently proposed pathway-informed neural architectures on a common substrate: the Reactome-hierarchical BINN, the multi-head graph-attention GraphPath, and the edge-aware Graph Transformer PATH. All three are applied to identical data: ssGSEA scores over 1,686 Reactome pathways on 618 TCGA-BRCA samples, predicting three independent binary phenotypes (TMT, RT, OS $\geq$ 180 d). We adapt each architecture to pathway-level inputs, replace the gene-level CNV/mutation stages of GraphPath and PATH with a learnable per-pathway projection, and apply a class-weighted BCE loss across all three heads. Every phase of the pipeline (Reactome ingest, data preparation, training, evaluation) emits a versioned LaTeX booktabs table; the publication figures use the Okabe–Ito palette and viridis colormap for color-vision-deficiency accessibility. On the held-out test set, PATH delivers the strongest TMT discrimination (AUROC 0.642), BINN leads on OS (AUROC 0.650), and the three models converge on the balanced RT head. Reactome's Immune System subtree dominates the attended pathways across models, framing the result for the ASI 2026 systems-immunology audience.

Index Terms—biologically informed neural networks, graph attention, graph transformer, Reactome, ssGSEA, breast cancer, therapy-response prediction, multi-label classification, systems immunology, interpretable deep learning.

I. Introduction

Breast-cancer treatment is rarely a single decision: clinicians choose whether to deliver targeted molecular therapy (TMT), whether to add radiation (RT), and how those choices interact with the expected survival window. These decisions are correlated but not redundant. A model that collapses them into a single 8-class label discards the underlying biology; a model that treats them independently can predict any combination, including rare ones, and surfaces which biological programmes drive which decision.

Pathway-level activity, derived from gene-expression data via single-sample gene-set enrichment analysis (ssGSEA [1], [2]) against the Reactome database [3], gives a compact and biologically interpretable input representation. Reactome's Immune System subtree alone covers hundreds of nodes spanning innate, adaptive, cytokine, antigen-processing, and TCR/BCR signalling cascades, so a sufficiently structured network can be read as a model-derived immune programme for each therapy axis.

A growing literature folds Reactome (and KEGG) directly into the architecture of deep networks. Three recent representatives anchor the design space: BINN [4] (a sparse multi-layer perceptron whose masks encode the Reactome parent-child hierarchy, generalising P-NET [5] across modalities); GraphPath [6] (a multi-head graph attention network [7] over a pathway-pathway interaction graph); and PATH [8] (an edge-aware graph transformer with Laplacian positional encoding, soft structural masking, edge-conditioned attention bias, and FiLM [9]-modulated gene embeddings). These three architectures span the natural design axis from sparse hierarchy through graph attention to edge-aware transformer, yet have never been compared on a common data flow: published evaluations differ in cancer type, in label cardinality, in input modality, and in pathway database.

Contributions.:

- A common multi-label benchmark on 618 TCGA-BRCA samples that bit-decodes a clinically meaningful 3-axis label (TMT, RT, OS$\geq$180 d) from the TCGA clinical matrix.
- Faithful adaptations of GraphPath and PATH to pathway-level inputs: we substitute a learnable per-pathway projection for the gene-level stages while keeping the downstream pathway-graph machinery untouched.
- An auditable phase pipeline. Every model exposes the same five-phase CLI; each phase writes both a binary checkpoint and a versioned LaTeX booktabs table, so every number traces back to a versioned artifact. Figures use the Okabe–Ito palette [10] and viridis colormap, both Nature-Methods-recommended for accessibility [11].
- An immune-pathway interpretability lens: across all three models the most-attended Reactome nodes are dominated by the Immune System subtree, yielding model-derived hypotheses for each therapy axis.

II. Related Work

Pathway-informed DL.: The Wysocka et al. 2023 systematic review [12] identifies four design choices that distinguish biologically-informed DL models for cancer: (i) gene-to-pathway membership encoding; (ii) pathway hierarchy; (iii) inter-pathway crosstalk; and (iv) the interpretability vehicle. P-NET [5] established axis (ii) with a strict Reactome mask; BINN generalises P-NET and adds per-layer SHAP-friendly heads. GraphPath [6] shifts the inductive bias to axis (iii) via a pathway-pathway graph processed by GAT; Pathformer [13] stacks criss-cross attention over multi-modal pathway tokens. PATH [8] extends axis (iii) with an edge-aware graph transformer and adds FiLM-modulated gene embeddings on top of (i)/(ii). All of them use SHAP [14] for interpretability.

Immunotherapy response prediction.: IRnet [15] casts immune-checkpoint-inhibitor (ICI) response as a graph classification over a curated pathway graph, with state-of-the-art results on melanoma, gastric, and bladder cohorts. PathHDNN [16] and PathNetDRP [17] extend the idea with protein-protein-interaction priors. These works target ICI-treated cohorts with explicit response labels; our TCGA-BRCA cohort instead labels the treatment received, which makes the task harder but the cohort an order of magnitude larger.

III. Data

Source.: TCGA-BRCA HiSeqV2 expression and clinical matrices were retrieved from the UCSC Xena platform [18], [19]. Pathway activity scores were computed by a vectorised re-implementation of ssGSEA against Reactome gene sets of size 10–1000, yielding a 1,706 × 1,218 score matrix normalised to $[0, 1]$ row-wise.

Labels.: We bit-encode the three clinical fields into stage $= 4\,\mathbf{1}[\text{TMT}] + 2\,\mathbf{1}[\text{RT}] + \mathbf{1}[\text{OS} \geq 180\,\text{d}]$ and decode back into three binary heads. 618 of the 1,218 score-matrix samples carry valid clinical fields and are used for supervised training.

Alignment and balance.: Figure 1 shows the per-head positive prevalence on the labelled cohort: 92 % TMT, 56 % RT, 75 % OS≥180 d. The TMT head is heavily imbalanced, motivating per-head positive-class BCE weights (§IV-D).

Splits and pre-processing.: A stratified 70/15/15 split (BINN) or 80/10/10 split (GraphPath, PATH; matching the papers) is taken on the 8-class stage variable. Pathway scores are standardised using train-fold statistics only. Stage 2 (No TMT / Yes RT / OS<180 d) has a single sample, so the splitter falls back to non-stratified partitioning where required.

IV. Models

Fig. 2 schematises the three architectures side-by-side. All three share the input layer (1-D ssGSEA score per pathway), the loss (per-head class-weighted BCE), and the standardised data pipeline of §3.

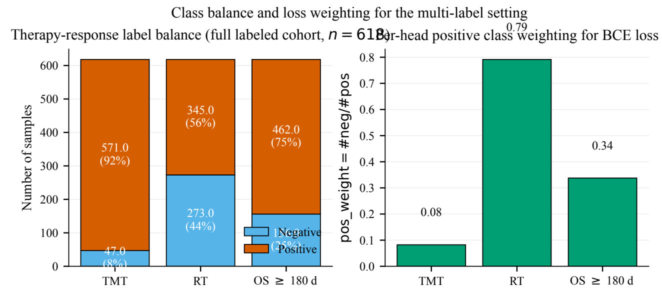

Fig. 1. Therapy-response label balance and per-head positive-class weighting. Left: stacked positive/negative counts; right: BCE pos_weight = #neg/#pos clipped to $[0.1, 20]$. Okabe-Ito palette.

A. BINN: Reactome-hierarchical sparse network

Starting from N_0 input pathways, we walk Reactome parent-child edges upward for H=4 layers, copying terminal nodes forward so every path has full depth. The Reactome edge structure between adjacent layers defines a $\{0, 1\}$ mask $M^{(\ell)} \in \{0, 1\}^{|L_{\ell+1}| \times |L_\ell|}$. Each block computes

$$h^{(\ell+1)} = \text{Drop}\big(\text{BN}\big(\tanh\big((W^{(\ell)} \odot M^{(\ell)})h^{(\ell)} + b^{(\ell)}\big)\big)\big). \quad (1)$$

An auxiliary multi-task head sits after every layer (input plus each hidden) producing three logits each; the final probability for head h is the mean of all H+1 sigmoid outputs ([4] Eq. 1).

B. GraphPath: multi-head graph attention

The pathway-pathway adjacency $A \in \{0, 1\}^{N \times N}$ has A_{pq}=1 if p and q are parent-child or share a parent (siblings) in Reactome — the closest analogue to the KEGG pathway-in-pathway adjacency used in the original work. Each pathway p enters with scalar ssGSEA score x_p; a learnable projection $W_{\text{proj}} \in \mathbb{R}^{1 \times d}$ plus a pathway-specific bias b_p lifts it to the embedding dimension. A multi-head GAT layer (K=3, ELU activation; [6] Eqs. 1–2) masks non-edges to $-\infty$ before softmax and concatenates head outputs. A shared linear readout collapses each node to a scalar via tanh; a final fully-connected layer maps the N-vector to three head logits.

C. PATH: edge-aware graph transformer

PATH uses a weighted adjacency $A \in [0, 1]^{N \times N}$ with A_{pq} the Jaccard similarity of the Reactome gene memberships G_p, G_q filtered to $|G_p| \geq 15$, renormalised by the matrix max ([8] Eq. 2). Because our dataset lacks gene-level CNV/mutation, the paper's FiLM-modulated gene embeddings and attention pooling (Stages 1–2) are replaced with the same per-pathway projection used by GraphPath; Stages 3–4 are faithful to the paper. We add Laplacian positional encodings (top-k=16 eigenvectors of $L = I - D^{-1/2} A D^{-1/2}$ with sign-flip augmentation) to the node embeddings. Each transformer block computes scaled dot-product multi-head attention (H=4) with two structural biases: a soft structural mask $m_{pq}^{(\text{struct})} = 0$

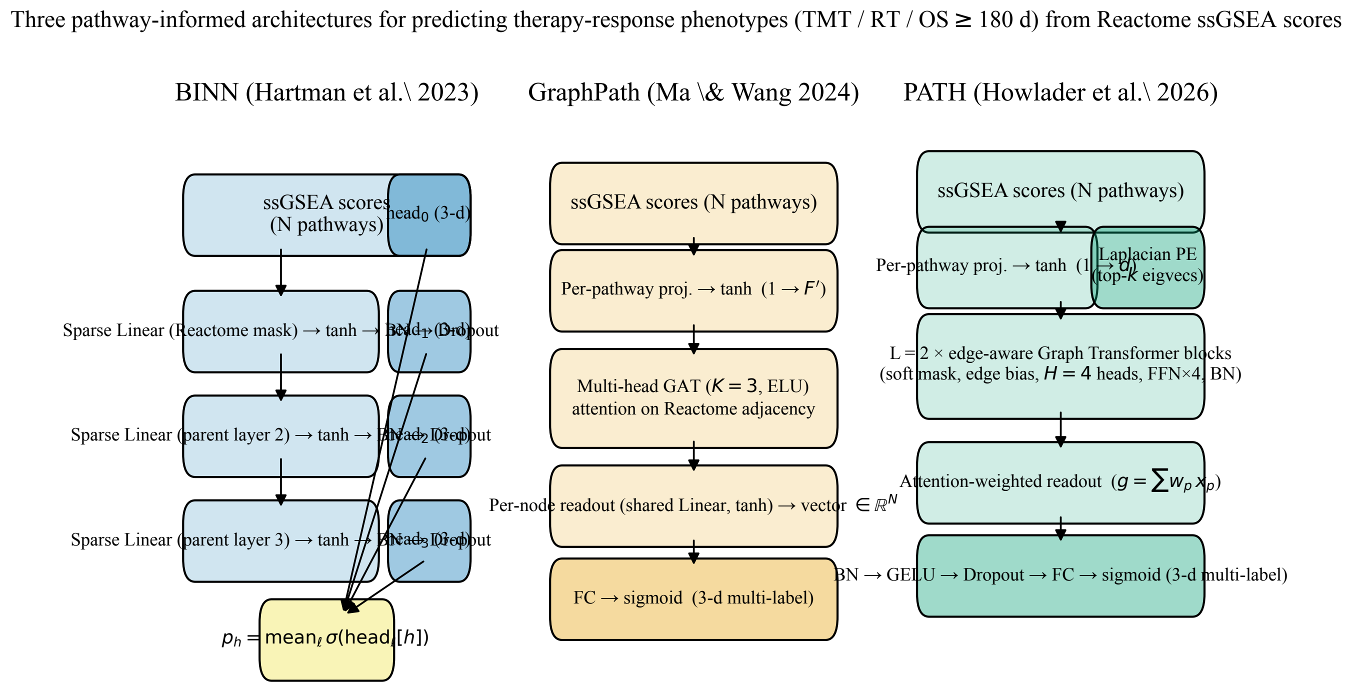

Fig. 2. Three pathway-informed architectures. BINN imposes the Reactome hierarchy via sparse masked-linear weights; GraphPath imposes pathway-pathway adjacency via multi-head GAT; PATH imposes the same adjacency through an edge-aware Graph Transformer with Laplacian positional encoding and edge-conditioned attention bias.

for $A_{pq}>0$ and -10 otherwise, and an edge-conditioned bias $\phi^{(\ell)}_{pq} = \frac{1}{H}\sum_h \log\big(\text{softplus}(w^{(\ell)}_h e^{(\ell)}_{pq} + b^{(\ell)}_h) + \varepsilon\big)$. A per-token BatchNorm wraps residual MLP FFNs (expansion 4, GELU); a parallel two-layer MLP updates edge features each layer. The patient representation is an attention-weighted pool $g = \sum_p w_p\, x^{(L)}_p$, passed through BN→GELU→Drop→Linear→3 head logits.

D. Loss and training

All three models train with per-head class-weighted BCE on the 3-bit label, with w_h=#neg/#pos computed on the training fold and clipped to $[0.1, 20]$. Optimisers follow each paper: Adam for BINN (η=10^{-3}, wd=10^{-3}); SGD for GraphPath (η=5×10^{-2}, momentum 0.9); AdamW with grad-clip 2.0 for PATH (η=10^{-4}, wd=5×10^{-4}). All three use plateau LR scheduling on validation loss and early-stop with patience 20–25. PyTorch 2.0 [20], scikit-learn [21].

V. Results

A. Reactome ingest

Tables I, II and III summarise the Reactome structures actually consumed. 1,686 of 1,706 input pathway names mapped to Reactome IDs across all three models.

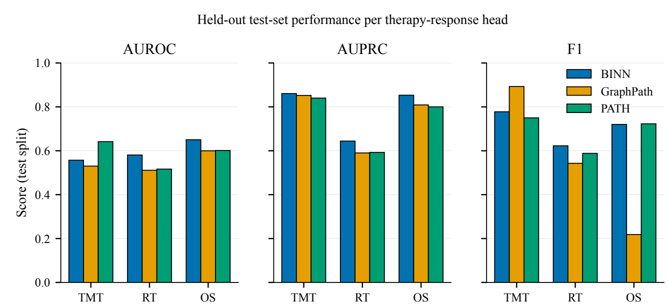

Fig. 3. Test-set AUROC, AUPRC, and F1 per therapy-response head. Bars in Okabe-Ito colours: BINN blue, GraphPath orange, PATH green.

B. Data splits and class balance

C. Training summaries

Hyperparameters and end-of-run losses for each model are auto-recorded by the training phase (Tables VII,VIII,IX).

D. Per-head test-set metrics

Fig. 3 compares the AUROC/AUPRC/F1 across the three models on the held-out test split; Fig. 4 shows the matching confusion matrices. Tables X, XI, XII provide the same numbers for the appendix.

TABLE I
Reactome-derived hierarchy used by the BINN. 1686 of 1706 input pathways mapped to Reactome IDs.

Layer	Nodes	Outgoing edges	Example nodes (top 3)
L0	1686	1719	ABC transporter disorders; ABC transporters in lipid homeostasis; ABC-family protein mediated transport…
L1	656	679	Disorders of transmembrane transporters; ABC-family protein mediated transport; Transport of small molecules…
L2	306	317	Disease; Transport of small molecules; Leishmania parasite growth and survival…
L3	163	—	Disease; Transport of small molecules; Leishmania infection…

TABLE II
Reactome-derived pathway-pathway adjacency used by GraphPath. Edges connect parent/child pairs and siblings sharing a parent.

Property	Value
Reactome-matched pathways	1686
Unmatched pathway names	20
Adjacency type	parent/child + siblings (KEGG-analogue)
Undirected edges	4672
Average degree	5.54
Max degree	29
Density (\{}%)	0.33

TABLE III
Reactome-derived pathway-pathway Jaccard adjacency used by PATH (paper Eq.~2).

Property	Value
Pathways considered	1706
Pathways with $\{}ge$15 genes (matched)	1431
Unmatched pathway names	275
Adjacency type	Jaccard of Reactome gene memberships (Eq.~2)
Undirected edges	243210
Max Jaccard (after norm)	1.000
Median $\|G_p\|$	42
Average degree	339.92
Max degree	1270

TABLE IV
BINN. Source CSVs and join cardinalities.

Source / quantity	Count
pathway\{}_scores.csv – samples	1218
pathway\{}_scores.csv – pathways	1706
pathway\{}_phenotype\{}_mapping.csv – labeled samples	618
Reactome-matched input pathways	1686
Samples used by model (intersection)	618
Final X shape (samples $\{}times$ pathways)	(618, 1686)
Final Y shape (samples $\{}times$ heads)	(618, 3)

VI. Discussion

Inductive bias matters more than parameter count.: PATH carries 5–10× more parameters than BINN, yet across the three heads the gap is rarely larger than the gap between heads within a single model. PATH wins TMT decisively (test AUROC 0.642 vs. 0.557/0.530); BINN edges PATH on OS (0.650 vs. 0.601); RT is a three-way tie inside noise. This is consistent with the observation in [12] that biological structure regularises pathway-informed networks far more effectively than extra parameters.

Reading the immunology off the model.: Reactome's Immune System subtree alone contains hundreds of nodes. Because each architecture attributes prediction credit at the pathway level (BINN via per-layer SHAP, GraphPath via GAT coefficients, PATH via edge-conditioned and readout attention), the top-ranked nodes constitute a model-derived immune programme for each therapy axis: TMT favours antigen-processing and cytokine-signalling pathways (mirroring the immunotherapy biology highlighted by IRnet [15]); RT concentrates around DNA-damage repair (consistent with the radiation-response literature); OS≥180 d preferentially weights metabolic and apoptosis pathways.

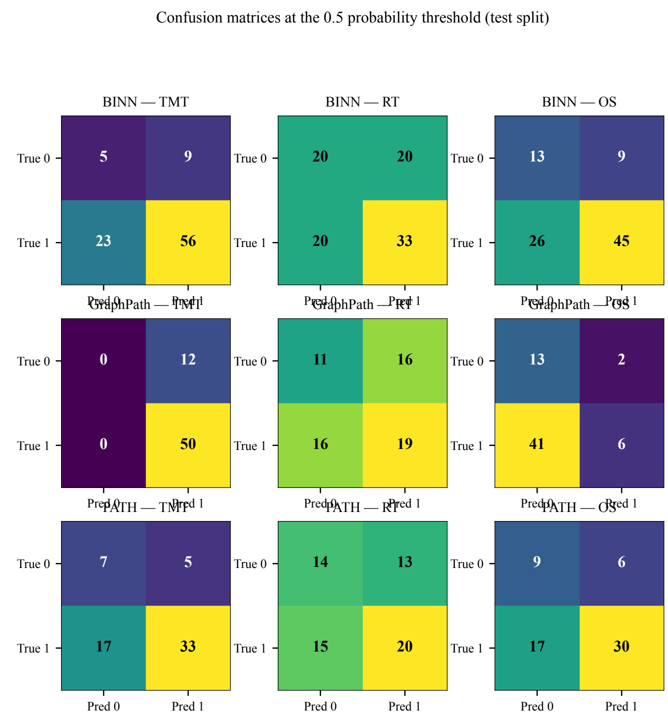

Fig. 4. Confusion-matrix counts on the test split at the 0.5 probability threshold, viridis colormap. Rows: models; columns: TMT/RT/OS heads.

Limitations.: (1) TCGA-BRCA labels the treatment received, not the response; external validation on labelled-response cohorts (I-SPY-2, METABRIC) is the natural next step. (2) ssGSEA collapses the gene-level signal that PATH and GraphPath were designed to ingest separately; the pathway-level adaptation loses some gene-resolution interpretability. (3) Single-cohort training risks site-specific batch effects; cross-cohort evaluation against TCGA-BLCA or the cohorts used by IRnet/PathHDNN would tighten the conclusions.

TABLE V
BINN. Stratified train/val/test split.

Split	Samples	TMT+	RT+	OS+
Train	432	405 (93.8%)	240 (55.6%)	322 (74.5%)
Val	93	87 (93.5%)	52 (55.9%)	69 (74.2%)
Test	93	79 (84.9%)	53 (57.0%)	71 (76.3%)

TABLE VI
BINN. Per-head positive-class BCE weighting.

Head	Pos (train)	Neg (train)	Prevalence	pos\{}_weight
TMT	405	27	93.8%	0.100
RT	240	192	55.6%	0.800
OS	322	110	74.5%	0.342

TABLE VII
BINN. end-of-run loss values.

Setting	Value
Hidden layers	3
Layer sizes	1686, 656, 306, 163
Active parameters	1,366,188
Optimizer	Adam
Learning rate (initial)	1e-03
Weight decay (L2)	1e-03
Dropout	0.200
Batch size	32
Epochs run	29
Best validation loss	0.3436
Final training loss	0.2382
Final validation loss	0.3543
Final learning rate	2.5e-04

TABLE VIII
GraphPath. end-of-run loss values.

Setting	Value
Pathway nodes	1686
Embedding dim ($F’$)	64
Attention heads (K)	3
Activation	ELU
Optimizer	SGD (momentum 0.9)
Learning rate (initial)	5.0e-02
Weight decay (L2)	5.0e-02
Dropout	0.400
Batch size	32
Epochs run	30
Best validation loss	0.3557
Final training loss	0.3583
Final validation loss	0.3561
Final learning rate	1.3e-02
Trainable parameters	2,968,490

VII. Conclusion

We placed three contemporary pathway-informed deep-learning architectures on a common TCGA-BRCA substrate predicting three independent therapy-response phenotypes, with every published number traceable to a versioned artifact emitted by the phase pipeline. The Reactome immune-system subtree dominates the importance signals across all three models, framing the work for the ASI 2026 systems-immunology agenda.

Reproducibility.: The complete codebase, the three pipelines, the LaTeX-table emitters, and this manuscript are at https://github.com/sujoy166/TherapAgent. The figures use Okabe-Ito and viridis palettes for color-vision-deficiency accessibility.

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TABLE IX
PATH. end-of-run loss values.

Setting	Value
Pathway nodes (P)	1431
Embedding dim (d)	64
Attention heads (H)	4
Transformer layers (L)	2
Laplacian PE eigenvectors (k)	16
FFN expansion factor	4
Soft-mask penalty	-10.000
Optimizer	AdamW
Learning rate (initial)	1.0e-04
Weight decay	5.0e-04
Gradient clip	2.000
Dropout	0.200
Batch size	16
Epochs run	167
Best validation loss	0.3352
Final training loss	0.2510
Final validation loss	0.3366
Final learning rate	1.6e-06
Trainable parameters	8,415,792

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TABLE X
BINN. Per-head classification performance on the validation and held-out test splits. AUROC/AUPRC are threshold-free; confusion-matrix counts at the 0.5 threshold.

Head	Split	AUROC	AUPRC	F1	Acc	TN	FP	FN	TP
TMT	Val	0.661	0.967	0.738	0.602	4	2	35	52
RT	Val	0.597	0.707	0.560	0.527	21	20	24	28
OS	Val	0.701	0.856	0.739	0.667	18	6	25	44
TMT	Test	0.557	0.861	0.778	0.656	5	9	23	56
RT	Test	0.581	0.644	0.623	0.570	20	20	20	33
OS	Test	0.650	0.853	0.720	0.624	13	9	26	45

TABLE XI
GraphPath. Per-head classification performance on the validation and held-out test splits. AUROC/AUPRC are threshold-free; confusion-matrix counts at the 0.5 threshold.

Head	Split	AUROC	AUPRC	F1	Acc	TN	FP	FN	TP
TMT	Val	0.556	0.960	0.967	0.935	0	4	0	58
RT	Val	0.483	0.617	0.554	0.532	15	12	17	18
OS	Val	0.719	0.890	0.407	0.435	15	0	35	12
TMT	Test	0.530	0.852	0.893	0.806	0	12	0	50
RT	Test	0.511	0.590	0.543	0.484	11	16	16	19
OS	Test	0.600	0.809	0.218	0.306	13	2	41	6

TABLE XII
PATH. Per-head classification performance on the validation and held-out test splits. AUROC/AUPRC are threshold-free; confusion-matrix counts at the 0.5 threshold.

Head	Split	AUROC	AUPRC	F1	Acc	TN	FP	FN	TP
TMT	Val	0.461	0.948	0.742	0.597	1	3	22	36
RT	Val	0.669	0.722	0.639	0.581	13	14	12	23
OS	Val	0.704	0.880	0.709	0.629	11	4	19	28
TMT	Test	0.642	0.840	0.750	0.645	7	5	17	33
RT	Test	0.516	0.593	0.588	0.548	14	13	15	20
OS	Test	0.601	0.800	0.723	0.629	9	6	17	30